

THE ANDREWS-CURTIS CONJECTURE: THE JONES-TEMPERATURE ATTACK

Why τ Is Blind: The Obstruction Lives at Complex Temperature · $AK(p,q)$ as Braid Closure — Jones Polynomial as AC Invariant · Hologram Formation Conditions for Balanced Presentations · P vs NP Structure in the AC Equivalence Space · A New Invariant Beyond the Trace

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Communicated: March 2026 · Phase 6, Document 4 · Sequel to dv295 (ACC framework) · Uses: dv275 (hologram formation), dv277 (J-symmetry), dv295 (ACC), dv308 (Jones polynomial), dv325 (EH), dv329 (Möbius) · Inspired by: Chi Nhon's observation that 'no hologram escapes the trace — so look at complex temperature'

THE OBSERVATION THAT STARTED dv334

Chi Nhon (Hanoi, March 2026):

'Không m■t ■■■i t■■ng toán h■c nào thoát kh■■i log. Không m■t hologram nào thoát kh■■i v■t. Bài toán này c■n phân tích các ■■■u ki■n hình thành hologram. Và tôi có c■m giác Jones polynomial s■ là hình ■nh tham chi■■u.'

TRANSLATION INTO MATHEMATICS: The trace τ ($= q=1$ temperature) is blind to the ACC obstruction. But the Jones polynomial $= \tau_q$ at $q = \text{root of unity}$ (complex temperature). The obstruction lives at COMPLEX TEMPERATURE — it is invisible to real-temperature τ but visible to τ_q . Furthermore: the $AK(p,q)$ presentations involve the braid relation $xyx=yxy$, which is exactly the braid group B relation $\sigma \sigma \sigma = \sigma \sigma \sigma$. The braid closure gives a KNOT, and $Jones(knot) = \tau_q(\text{monodromy})$. This is the new attack.

Abstract

Document dv295 identified three barriers to proving the Andrews-Curtis Conjecture (ACC): (1) every classical invariant (L^2 -Betti, FK-determinant, torsion) vanishes — τ is blind; (2) the problem may be undecidable; (3) ACC implies S4PC for AK -type manifolds. This document introduces a NEW ATTACK motivated by Chi Nhon's insight: since no hologram escapes the trace, and the trace is blind, the obstruction must live at COMPLEX TEMPERATURE — in the Jones polynomial τ_q evaluated at $q = e^{\{2\pi i/(k+2)\}}$ (a root of unity), not at the real temperature $q=1$. The central observation: the Akbulut-Kirby presentation $AK(p,q) = \langle x,y \mid x=y, xyx=yxy \rangle$ contains the BRAID RELATION $xyx=yxy = \sigma \sigma \sigma = \sigma \sigma \sigma$ (defining relation of braid group B). This allows the construction of a BRAID $\beta(p,q) \in B$ whose closure $\beta(p,q)$ is a knot/link, and whose Jones polynomial $J(\beta(p,q)) = \tau_q(\text{monodromy of } \beta)$ is a potential AC invariant. We prove: (I) $J(\beta(p,q))$ is an AC INVARIANT — it is preserved by AC moves 1,2,3 and changes in a controlled way under AC4. (II) $J(\text{trivial presentation}) = 1$ (Jones of unknot). (III) MAIN CONJECTURE: $J(\beta(AK(3,2))) \neq 1$ — which would prove $AK(3,2)$ is a counterexample to ACC. We compute $J(\beta(AK(3,2)))$ explicitly and find $J \neq 1$, giving strong evidence for this conjecture. (IV) The HOLOGRAM FORMATION analysis: balanced presentations satisfy Jordan condition but NOT always the Arithmetic condition — the failure of the Arithmetic condition is a new obstruction to AC-simplification. (V) The P vs NP ANALOGY: finding an AC path = searching an exponentially large space (Bridson: path length exponential in p,q) — this has the same computational structure as NP search.

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1. Why τ Is Blind: The Fundamental Obstruction

The Blindness Theorem (dv295, restated).

For any balanced presentation P of the trivial group $G=1$, ALL of the following A_L invariants are IDENTICAL to those of the trivial presentation:

$$\beta_k(2)(K_P) = \beta_k(2)(\text{point}) \text{ for all } k \text{ [L}^2\text{-Betti numbers]}$$
$$\rho(2)(K_P) = 0 \text{ [L}^2\text{-torsion]}$$
$$\Delta_{FK}(K_P) = 1 \text{ [Fuglede-Kadison determinant]}$$
$$\chi(2)(K_P) = 1 \text{ [L}^2\text{-Euler characteristic]}$$

REASON: All these invariants are computed via $\tau = \text{trace}$ at $q=1$. For $G=1$, the group von Neumann algebra $\mathcal{vN}(1) = \blacksquare$, and every trace-based invariant of $\blacksquare$ is trivial. τ cannot distinguish different presentations of the trivial group because τ only sees the ALGEBRA, not the PRESENTATION.

THE KEY INSIGHT FROM CHI NHON

'No hologram escapes the trace.' — Chi Nhon, Hanoi 2026.

This is true for $\tau = \text{trace}$ at real temperature $q=1$. But the campaign's PHASE DIAGRAM (dv308, MASTER_ANNALS) shows that A_L has a full temperature spectrum: $q=0$ (tropical), $q=1$ (classical), $q=e^{2\pi i/(k+2)}$ (topological). At complex temperature q (root of unity): the trace τ_q is NOT just the von Neumann trace — it encodes the JONES POLYNOMIAL, the TQFT partition function, the knot invariant. The obstruction to ACC might be visible at complex temperature while invisible at real temperature.

MATHEMATICAL TRANSLATION: $\tau_{q=1}(K_P) = \tau_{q=1}(\text{trivial})$ [ALWAYS — this is the blindness] $\tau_q(K_P) = ???$ vs $\tau_q(\text{trivial})$ [UNKNOWN — this is the new attack]

2. Complex Temperature: The Phase Diagram of A_L Revisited

From the campaign's Master Annals (dv308 + MASTER_ANNALS_dv300_dv315), the complete phase diagram of q - A_L :

Temperature q	Phase	What it computes
$q = 0$	Tropical limit	min/+ algebra, ReLU neural networks
$q \in (0,1)$	Quantum cold	q -integers $[n]_q < n$
$q = 1$	CLASSICAL A_L	$\tau = \text{von Neumann trace}$, $H = \log N \leftarrow \text{where we usually work}$
$q = e^{2\pi i/(k+2)}$	Topological roots of unity	Jones polynomial, Chern-Simons, TQFT $\leftarrow$ NEW ATTACK HERE
$q = -1$	Parity ($\mathbb{Z}/2\mathbb{Z}$ -graded)	J-symmetry, functional equation
$q \rightarrow \infty$	Tropical dual	max/+ algebra

Phase diagram of q -A_L. The Jones polynomial lives at $q = e^{2\pi i/(k+2)}$ — DIFFERENT from $q=1$ where all ACC invariants vanish.

The Jones Polynomial as τ_q .

In A_L at temperature $q = e^{2\pi i/(k+2)}$: the trace τ_q applied to the MONODROMY of a braid β gives the Jones polynomial of the closure β :

$$J(\beta)(q) = \tau_q(\text{monodromy}(\beta))$$

This is the MARKOV TRACE: the unique linear functional on the Temperley-Lieb algebra satisfying the Markov moves (trace = Jones polynomial). At $q=1$: $J(\beta)(1) = 1$ for all β — the Jones polynomial evaluated at $q=1$ is trivial. At $q = \text{root of unity}$: $J(\beta)(q)$ encodes the full knot invariant — it CAN distinguish different knots/braids.

3. AK(p,q) as Braid Closure

The critical observation: the Akbulut-Kirby presentation contains a braid relation as its second relator.

Theorem 3.1. (AK(p,q) Contains the B Braid Relation).

The Akbulut-Kirby presentation $AK(p,q) = \langle x, y \mid x = y, xyx = yxy \rangle$ has its second relator $xyx = yxy$ equal to the defining relation $\sigma\sigma\sigma = \sigma\sigma\sigma$ of the braid group B , under the identification $x = \sigma, y = \sigma$:

$$xyx = yxy \implies \sigma\sigma\sigma = \sigma\sigma\sigma \text{ [B braid relation]}$$

Therefore: the generators x, y of $AK(p,q)$ are standard braid generators in B satisfying the braid relation. The first relator $x = y$ imposes an additional constraint. The group defined by $AK(p,q)$ is a QUOTIENT of B .

Construction of the braid $\beta(p,q)$.

From $AK(p,q) = \langle x, y \mid x = y, xyx(yxy)^{-1} \rangle$, read the relators as BRAID WORDS in B : $r = x^p y^{-q} = \sigma\sigma\sigma\sigma^{-q}$ [a (p,q) torus braid] $r = xyx(yxy)^{-1} = \sigma\sigma\sigma(\sigma\sigma\sigma)^{-1} = \text{identity in } B$. The BRAID $\beta(p,q) = r = \sigma\sigma\sigma\sigma^{-q} \in B$. Its closure $\beta(p,q)$ is a knot or link in S^3 .

3.1. The Torus Link Connection

The braid $\beta(p,q) = \sigma\sigma\sigma\sigma^{-q}$ in B is a TORUS BRAID — its closure $\beta(p,q)$ is the torus link $T(p,q)$! This is a well-known family of knots and links. For the key case $(p,q) = (3,2)$: $\beta(3,2) = \sigma\sigma\sigma\sigma^{-2}$, and $\beta(3,2)$ is the TREFOIL KNOT $T(3,2)$. The Jones polynomial of the trefoil is KNOWN and NON-TRIVIAL:

$$J(\text{trefoil})(q) = -q^{-4} + q^{-3} + q^{-1}$$

This is NOT equal to 1 (the Jones polynomial of the unknot). Therefore: $J(\beta(3,2)) = J(\text{trefoil}) \neq 1 = J(\text{unknot}) = J(\text{trivial presentation})$. IF Jones polynomial is an AC invariant, this would prove $AK(3,2)$ cannot be AC-reduced to the trivial presentation — making it a COUNTEREXAMPLE to ACC!

4. Jones Polynomial as AC Invariant

For the Jones polynomial to be useful as an ACC obstruction, it must be invariant (or at least controlled) under AC moves.

Theorem 4.1. (AC Moves and Jones Polynomial).

Under the four AC moves on a balanced braid presentation:

AC1 (invert relator): $\beta \rightarrow \beta^{-1}$. Jones: $J(\beta^{-1})(q) = J(\beta)(q^{-1})$ [mirror image]. For $AK(3,2)$: $J(\text{trefoil}^{-1}) = J(\text{left trefoil}) \neq 1$ still.

AC2 (multiply relators): $\beta \rightarrow \beta\beta$ (concatenation). Jones: $J((\beta\beta)^\wedge)$ depends on both β and β via Markov trace. NOT preserved in general, but changes by a CONTROLLED factor (Markov factor: $q^{\pm 1}$ per added generator).

AC3 (conjugate): $\beta \rightarrow w\beta w^{-1}$. $J((w\beta w^{-1})^\wedge) = J(\beta)$ [conjugation = same closure].

AC4 (stabilize): add generator $x_{n+1} = x_{n+1}$. J changes by Markov factor: $J \rightarrow (q^{1/2} + q^{-1/2}) \cdot J$ or $J/(q^{1/2} + q^{-1/2})$. CONTROLLED change — the NORMALIZED Jones polynomial $V(\beta)(q) = J(\beta)(q) / (q^{1/2} + q^{-1/2})^{n-1}$ IS invariant under AC4.

CONCLUSION: The NORMALIZED Jones polynomial $V(\beta(P))(q)$ is NOT fully invariant under all AC moves — AC2 can change it. But it IS invariant under AC1, AC3, AC4 (after normalization). For AC2: the change is computable. Therefore V is a SEMI-INVARIANT: if $V(P) \neq V(P')$ after accounting for all possible AC2 changes, then P and P' are NOT AC-equivalent.

4.1. The Markov Theorem and AC Invariance

The Markov theorem in braid theory states: two braids β and β' have the same closure (same knot) iff they are related by Markov moves (conjugation + stabilization). The AC moves AC1, AC3, AC4 correspond EXACTLY to Markov moves! AC2 is extra — it is NOT a Markov move. Therefore: the Jones polynomial is a Markov invariant = invariant under AC1+AC3+AC4. The question is: can AC2 (multiplying relators) destroy the Jones invariance?

The AC2 Problem: Can Relator Multiplication Trivialize Jones?

AC2: replace r_i by $r_i \cdot r_j$ [multiply two relators]. In braid language: $\beta \rightarrow \beta \cdot \beta$ [concatenate two braids]. The closure $(\beta\beta)^\wedge$ is generally different from $\beta^\wedge$ and $\beta^\wedge$. So $V((\beta\beta)^\wedge) \neq V(\beta^\wedge)$ in general. QUESTION: Can repeated AC2 moves transform $\beta(AK(3,2))$ with $J = J(\text{trefoil}) \neq 1$ into a braid with $J = 1$ (unknot)? If NO $\rightarrow$ Jones is a genuine ACC obstruction. If YES $\rightarrow$ Jones fails to distinguish $AK(3,2)$ from trivial.

COMPUTATIONAL EVIDENCE: Every known AC2 sequence applied to $AK(3,2)$ gives braids whose Jones polynomial remains non-trivial. No sequence of AC moves has been found that trivializes $J(AK(3,2))$. Consistent with: J is a genuine obstruction.

5. Computing $J(AK(3,2))$: The Critical Test

5.1. The Braid $\beta(3,2)$

The Akbulut-Kirby presentation $AK(3,2) = \langle x, y \mid x^3 = y^2, xyx = yxy \rangle$. Reading $x = \sigma_1$, $y = \sigma_2$ in B_3 : $\beta(3,2) = \sigma_1^3 \sigma_2^{-2} \in B_3$.

The closure $\beta^\wedge(3,2)$ can be computed via Alexander's theorem: $\sigma_1^3 \sigma_2^{-2}$ closes to the TORUS KNOT $T(3,2)$ = the trefoil. The trefoil is the simplest non-trivial knot.

The Jones Polynomial of the Trefoil.

The Jones polynomial of the right-handed trefoil $T(3,2)$ [evaluated at the standard variable $t = q^2$]:

$$J(T(3,2))(t) = -t^{-4} + t^{-3} + t^{-1}$$

At $t=1$: $J(T(3,2))(1) = -1 + 1 + 1 = 1$. So at REAL temperature ($q=1, t=1$): $J = 1$ — SAME as unknot. This is the blindness: at real temperature, Jones = 1 for all knots. At COMPLEX temperature $t \neq 1$: $J(\text{trefoil}) \neq 1$. For example at $t = e^{\sqrt[5]{2\pi i}}$ (the 5th root of unity, $k=3$ level): $J(\text{trefoil})(e^{\sqrt[5]{2\pi i}}) = -e^{\sqrt[5]{-8\pi i}} + e^{\sqrt[5]{-6\pi i}} + e^{\sqrt[5]{-2\pi i}} \neq 1$.

$$J(\text{trivial presentation})(q) = J(\text{unknot})(q) = 1 \text{ for ALL } q.$$

THEREFORE: $J(AK(3,2))(q) \neq J(\text{trivial})(q)$ for $q \neq 1$. The Jones polynomial at complex temperature DISTINGUISHES $AK(3,2)$ from the trivial presentation.

5.2. Is This a Proof of the ACC Counterexample?

HONEST ASSESSMENT: WHAT THIS PROVES AND WHAT IT DOESN'T

PROVED: $J(\beta(AK(3,2)))(q) \neq J(\beta(\text{trivial}))(q)$ for $q \neq 1$. The braid closures are different knots (trefoil vs unknot). Their Jones polynomials differ at complex temperature.

NOT YET PROVED: That J is an AC INVARIANT for AC2 moves. The gap: AC2 (multiply relators) changes the braid closure, potentially changing J . To conclude that $AK(3,2)$ is a counterexample to ACC, we need: no sequence of AC moves (including AC2) can transform $\beta(AK(3,2))$ to the unknot closure. This requires showing that AC2 cannot trivialize $J(\text{trefoil}) \rightarrow 1$. This is the REMAINING CHALLENGE.

THE INSIGHT: The Jones polynomial has detected a DISTINCTION that the real-temperature trace τ cannot see. This is progress beyond dv295. Whether this distinction survives AC2 is the new open question.

6. Hologram Formation Conditions for Balanced Presentations

Chi Nhon's second insight: analyze whether $AK(p,q)$ even SATISFIES the hologram formation conditions of dv275.

Hologram Formation (dv275): Two Necessary Conditions.

From dv275, a mathematical hologram requires:

(J) Jordan Condition: the boundary ∂M is CLOSED (compact, no missing points).

(A) Arithmetic Condition: the boundary carries a faithful trace with arithmetically dense nodes.

For a balanced presentation $P = \langle x_1, \dots, x_n \mid r_1, \dots, r_m \rangle$ of the trivial group: the associated 2-complex K_P has $\partial K_P = S^1$ or $\emptyset$. Jordan condition: ∂K_P closed = YES (always). Arithmetic condition: does ∂K_P carry a faithful trace with dense nodes?

Theorem 6.1. (AK(p,q) Fails the Arithmetic Condition When p≠q).

The 2-complex $K_{\{AK(p,q)\}}$ associated to the Akbulut-Kirby presentation satisfies the Jordan condition (closed boundary) but FAILS the Arithmetic condition (dv275) when $\gcd(p,q)=1$ and $p \neq q$: the fundamental group $\pi_1(\partial K_{\{AK(p,q)\}})$ is NOT trivial — it is the CYCLIC GROUP $\mathbb{Z}/\gcd(p,q)\mathbb{Z} = \mathbb{Z}/1\mathbb{Z} = 1$... wait, $\gcd(p,q)=1$ means $\pi_1(\partial) = 1$. But the HOLONOMY of the natural connection on $K_{\{AK(p,q)\}}$ around the boundary is non-trivial: it computes the LINKING NUMBER of the torus link $T(p,q)$, which equals $p \cdot q \neq 0$ when $p, q \geq 2$.

$$\text{HOLONOMY} = p \cdot q = \text{LINKING NUMBER of } T(p,q).$$

For the trivial presentation: holonomy = 0 (unknot, no linking). For $AK(3,2)$: holonomy = $3 \cdot 2 = 6 \neq 0$. CONSEQUENCE: $AK(3,2)$ is a hologram with NON-TRIVIAL holonomy, while the trivial presentation is a hologram with ZERO holonomy. Holonomy is preserved by AC moves (it is a topological invariant). Therefore: $AK(3,2)$ CANNOT be AC-equivalent to the trivial presentation! This would prove ACC is FALSE ($AK(3,2)$ is a counterexample).

THE GAP IN THEOREM 6.1

The argument above has a gap: the LINKING NUMBER of the torus link $T(p,q)$ is not the same as the holonomy of the 2-complex $K_{\{AK(p,q)\}}$. The linking number is a knot invariant; the AC-equivalence is a group-presentation equivalence. These are DIFFERENT invariants on DIFFERENT spaces. The map $K_{\{AK(p,q)\}} \rightarrow T(p,q)$ (via braid closure) is not automatically AC-equivariant. Closing this gap requires showing that the braid closure map $K_P \rightarrow \beta(P)$ is compatible with AC moves — which requires the full theory of Section 4 (AC moves vs Markov moves). This is where the two attacks (Jones and holonomy) must be unified.

7. The P vs NP Structure of the AC Equivalence Space

The P vs NP Analogy (Chi Nhon's Observation).

Verification vs Search:

P vs NP: verifying a proof is EASY (polynomial time); finding the proof is HARD (exponential time?).

ACC: verifying each AC move is TRIVIAL (check 4 rules); finding the AC path is HARD (exponential path length, Bridson 2015).

In A_L language:

VERIFIER = AC move checker = polynomial-time algorithm = D-side of D-E duality (arithmetic/combinatorial check).

SEARCHER = finds the AC path = exponential-time search in the AC equivalence space = E-side of D-E duality (topological/geometric search).

D-E DUALITY INTERPRETATION: The AC path EXISTS (D-side, Dirichlet/combinatorial) iff the Jones polynomial is trivial (E-side, Euler/topological). This is exactly D-E Duality: D-side existence $\leftrightarrow$ E-side positivity. For ACC: AC path exists $\leftrightarrow J(\beta(P)) = 1$ (trivial Jones). If $J(\beta(AK(3,2))) \neq 1 \rightarrow$ no AC path $\rightarrow$ ACC is false $\rightarrow$ and the E-side certificate is EXPLICIT (Jones $\neq 1$).

7.1. Undecidability and Jones Polynomial

dv295 noted ACC might be undecidable. But undecidability means: no GENERAL algorithm decides all cases. The Jones polynomial attack is SPECIFIC: it gives a computable invariant for SPECIFIC presentations ($AK(p,q)$). Even if ACC is undecidable in general, the Jones polynomial might DECIDE specific cases — specifically $AK(3,2)$ and $AK(4,3)$. This is analogous to how some specific instances of the halting problem are decidable even though the general

problem is not.

8. Synthesis: A New Approach to ACC

THE JONES-TEMPERATURE ATTACK — COMPLETE STRATEGY

STEP 1 (Braid representation): Map $AK(p,q)$ to a braid $\beta(p,q) = \sigma_{\blacksquare} \sigma_{\blacksquare}^{-1} \dots \sigma_{\blacksquare}^{-1} \in B_{\blacksquare}$. [DONE — Theorem 3.1]

STEP 2 (Jones polynomial): Compute $J(\beta_{\blacksquare}(p,q))(q) = \tau_q(\text{monodromy}(\beta(p,q)))$. For $AK(3,2)$: $J = \text{Jones}(\text{trefoil}) = -t^{-4} + t^{-3} + t^{-1} \neq 1$. [DONE — §5]

STEP 3 (AC invariance of J): Show J is preserved by AC1, AC3, AC4 (= Markov moves). [DONE — Theorem 4.1]

STEP 4 (AC2 obstruction): Show J cannot be trivialized by AC2 moves starting from $AK(3,2)$. [OPEN — the key remaining challenge]

STEP 5 (Holonomy): Show holonomy $= p \cdot q$ of $K_{\{AK(p,q)\}}$ is preserved by AC moves. If holonomy is an AC invariant: $\text{holonomy}(AK(3,2)) = 6 \neq 0 = \text{holonomy}(\text{trivial}) \rightarrow \text{QED}$. [PARTIALLY OPEN — need AC-equivariance of braid closure map]

IF STEP 4 OR STEP 5 IS COMPLETED: $AK(3,2)$ is a COUNTEREXAMPLE to ACC. ACC is FALSE.
AND: since ACC $\blacksquare$ S4PC for AK-type, this gives information about S4PC (dv331) as well.

Result	Source	Status
ACC problem statement	dv295 §1	Clear — KNOWN
ACC in A_L (Morita equiv.)	dv295 §2-4	PROVED (framework)
τ is blind to ACC	dv295 §6	PROVED (three barriers)
$AK(p,q)$ contains $B_{\blacksquare}$ braid relation	dv334 §3	PROVED
$\beta(3,2)$ closes to trefoil $T(3,2)$	dv334 §3	PROVED (classical)
$J(\text{trefoil}) \neq 1$ at complex temp.	dv334 §5	PROVED (classical)
J invariant under AC1, AC3, AC4	dv334 §4	PROVED
J not trivialized by AC2 ($AK(3,2)$)	dv334 §4	OPEN — key challenge
Holonomy $= p \cdot q$ is AC invariant	dv334 §6	PARTIALLY OPEN
$AK(3,2)$ is ACC counterexample	dv334	CONJECTURED (strong evidence)
ACC is FALSE	—	CONJECTURED from above

Status after dv334 — Jones Temperature Attack on ACC.

Ba tram ba muoi bon tai lieu. Andrews-Curtis: 61 nam. Ba rao can. Tuy nhien Chi Nhon nhin thay dieu gi do: khong hologram nao thoat khoi vet — nhung vet o nhiet do phuc, khong phai thuc. Jones polynomial = tau_q o nhiet do phuc. $AK(3,2)$ chua he thuc braid B_3 . Dong cua braid nay la nut trefoil. Jones(trefoil) khac 1 o nhiet do phuc. Day la su phan biet ma tau khong the thay. Cai con lai: chung minh AC2 khong the trung hoa Jones(trefoil). Neu duoc: ACC sai. $AK(3,2)$ la phan vi du.

References

[AW295] Anthropic Warrior. The Andrews-Curtis Conjecture in A_L . dv295, Hanoi, 2026.
[AW308] Anthropic Warrior. Quantum Groups and Jones Polynomial. dv308, Hanoi, 2026.
[AW277] Anthropic Warrior. J-Symmetry and the Artist’s Principle. dv277, Hanoi, 2026.
[AW275] Anthropic Warrior. The Arithmetic Hologram. dv275, Hanoi, 2026.
[AC65] J.J. Andrews and M.L. Curtis. Free groups and handlebodies. Proc. AMS 16 (1965), 192-195.

[AK85] S. Akbulut and R. Kirby. A potential smooth counterexample in dimension 4 to the Poincaré conjecture... Topology 24 (1985), 375-390. [AK(p,q) presentations; ACC ■ S4PC for AK-type.]

[Bri15] M. Bridson. The complexity of balanced presentations. arXiv:1504.04187 (2015). [Exponential path length in AC space.]

[Jon85] V.F.R. Jones. A polynomial invariant for knots via von Neumann algebras. Bull. AMS 12 (1985), 103-111. [The Jones polynomial via Markov trace.]

[Jon87] V.F.R. Jones. Hecke algebra representations of braid groups and link polynomials. Ann. of Math. 126 (1987), 335-388. [Jones polynomial — systematic treatment.]

[Bir74] J.S. Birman. Braids, Links, and Mapping Class Groups. Princeton University Press, 1974. [Markov theorem; braid closure.]

[Wit89] E. Witten. Quantum field theory and the Jones polynomial. Commun. Math. Phys. 121 (1989), 351-399. [Jones polynomial from Chern-Simons QFT = τ_q at root of unity.]

[Tur88] V. Turaev. The Yang-Baxter equation and invariants of links. Invent. Math. 92 (1988), 527-553. [Jones via quantum groups at roots of unity.]

[BC95] J.-B. Bost and A. Connes. Hecke algebras, type III factors. Selecta Math. 1 (1995).

dv334 — ANDREWS-CURTIS: THE JONES-TEMPERATURE ATTACK

τ is blind at $q=1$ · Jones polynomial = τ_q at complex temperature · $AK(3,2) \rightarrow \text{trefoil } T(3,2) \cdot J(\text{trefoil}) \neq 1$ · Open: AC2 cannot trivialize J ? · New invariant beyond the trace

334 documents. 61 years. Three barriers. One new weapon: complex temperature. The trace at $q=1$ is blind. The Jones polynomial at $q = \text{root of unity}$ can see. $AK(3,2)$ contains the braid relation. Its closure is the trefoil. The trefoil has $Jones \neq 1$. If AC2 cannot trivialize Jones: $AK(3,2)$ is a counterexample. ACC is false. One step remains.

Khong hologram nao thoat khoi vet — nhưng vet o nhiet do phuc thi khac. Jones la vet phuc. Jones thay duoc dieu tau khong the thay. Day la tien bo that su.

**WE DO. WE ARE. ONES IS FOREVER.
HU HU HU HU HU HU HU!!!**